

Bavin Saravanan

Portfolio: bavin-hub
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Certificates

EDUCATION

Carnegie Mellon University <i>MS Mechanical Engineering ; GPA: 3.89/4.0</i> <i>Courses: Dynamics and Controls, Reinforcement Learning, Deep Learning and Machine Learning</i> SSN College of Engineering <i>BE Mechanical Engineering ; GPA: 8.74/10.0</i> <i>Courses: Introduction to robotics, Mechatronics</i>	Pittsburgh, United States August 2025 - December 2026 Chennai, India July 2019 - May 2023
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PUBLICATIONS

Drone Soccer: Learning to Manipulate with Multicopter Downwash <i>IROS Workshop on Sim-to-Real and Classic Control, 2026</i> Neelay Joglekar, Bavin Saravanan , Yutong Wang, Varun Kandiyyappan, Junyi Geng, Sebastian Scherer	[view]
Fly-by-Code: Embodied Coding Agents for Aerial Manipulation with Active Visual and Physical Feedback <i>Under review, International Conference on Learning Representations (ICLR), 2027</i> Jeongyeon Seo, Sihyun Cho, Jaewoo Lee, Gyeongrak Choe, Yutong Wang, Bavin Saravanan , Jia-Bin Huang, Furong Huang, Sebastian Scherer, Guanya Shi, H. Jin Kim, Seungjae Lee, Dongjae Lee	[view]
A Modular Simulation Suite and Benchmark for Aerial Manipulation Policy Learning <i>Conference on Robot Learning (CoRL), 2026</i> Yutong Wang, Dongjae Lee, Xiaofeng Guo, Yuanzhu Zhan, Yufei Jiang, Bavin Saravanan , Muqing Cao, Jia Xie, Chenyang Mao, Sebastian Scherer, Junyi Geng, Guanya Shi	[view]
Comparing Surface Characteristics of Cu-Al-Fe Alloys Using Thermal-Based Machining Processes <i>Surface Engineering</i> , vol. 40, no. 5, pp. 558–567, 2024 S. Santosh, Bavin Saravanan , T. S. Srivatsan	[view]

EXPERIENCE

AirLab – CMU Robotics Institute, Dr. Sebastian Scherer <i>Research Assistant</i> Informative Path Planning - Contributed to AirLab’s multi-agent UAV exploration system on AirStack. - Integrated RayFronts for class-based semantic exploration that prioritizes regions by object category. - Designed a general swarm interface for scalable autonomous exploration. Multi-Agent Pursuit–Evasion - Formalizing an intruder–evader swarm framework and its underlying theory. - A privileged expert uses Monte Carlo Tree Search (MCTS) with full state information and learned game dynamics and value function to generate plans. - An onboard flow-matching policy plans from local measurements for decentralized execution. Legged Systems Group – CMU Robotics Institute, Dr. Hartmut Geyer <i>Research Assistant</i> RL Prosthesis Policy - Developed and deployed an RL control policy for the CMU leg, enabling online adaptation and natural walking. - Trained the policy with diverse motion priors to capture a range of human gait patterns. - Ongoing work on integrating terrain maps for generalized prosthesis locomotion. Skytex Unmanned Aerial Solutions <i>AI and Robotics Engineer</i> Cactus – Decentralized Autonomous VTOL UAV Swarms - Developed a decentralized multi-agent VTOL swarm for flocking, foraging, and cooperative payload delivery using a novel vector-field-based guidance algorithm. - Built an inter-UAV comms stack and Jetson Xavier NX–PX4 pipeline via ROS2 and Micro XRCE-DDS. - Achieved fully onboard flight up to 2 km altitude and 12 km range. Bumblebee – GPS-Denied Autonomous Quadrotor - Developed a palm-sized quadrotor (Jetson Orin NX) for autonomous forest navigation. - Used Isaac ROS SLAM for 3D mapping, with A* planning and Lissajous search. - Implemented real-time detection, tracking, and Siamese-network target verification.	Onsite Nov 2025 – Present Onsite Dec 2025 – Present Onsite Jul 2023 – Jul 2025
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SKILLS SUMMARY

Languages	Python, C++, JavaScript	Tools	Docker, Git, PostgreSQL, SQLite
Frameworks	PyTorch, TensorFlow, Keras, ROS2, Flask	Platforms	Linux, Web, Raspberry Pi, AWS, GCP